ENERGY EFFICIENCY MODULE-HEAT EXCHANGERS

HEAT EXCHANGER PERFORMANCE ASSESSMENT

Heat Exchangers can deteriorate over time due to off-design operation, fouling, scaling, or corrosion.

This tool periodically assesses thermal duty, heat transfer coefficients (U), effectiveness (ϵ), and hydraulic pressure drops (ΔP) for both Shell & Tube and Plate Exchangers.

Metadata & Operational Configuration

Select Heat Exchanger Type:

Plate Heat Exchanger (PHE)

Design Data Inputs (from Test/Spec Reports)

Flow Configuration:

Counterflow

No. of Hot Side Passes

1

No. of Cold Side Passes

1

Total Design Heat Duty (kW)

6139.53

Overall Heat Transfer Coefficient U_d (Design/Service) (kW/m²-K)

7.2350

Design Fouling Margin (%)

Hot Fluid Side Pressure Drop Design (bar)

0.65

Cold Fluid Side Pressure Drop Design (bar)

0.98

Hot Fluid Temp Range (°C)

19.10

Cold Fluid Temp Range (°C)

16.90

Corrected LMTD Design (°C)

5.00

15.0

Operation Data Inputs

Hot Fluid Flow Inlet (kg/hr)
250000.00

Hot Fluid Temp Inlet (°C)
56.10

Hot Fluid Pressure Inlet (bar g)
4.50

Cold Fluid Flow Inlet (kg/hr)
290000.00

Hot Fluid Temp Outlet (°C)
39.00

Hot Fluid Pressure Outlet (bar g)
3.30

Cold Fluid Temp Inlet (°C)
32.00

Cold Fluid Pressure Inlet (bar g)
3.50

Cold Fluid Temp Outlet (°C)
43.00

Cold Fluid Pressure Outlet (bar g)
2.20

Fluid Properties

Hot Fluid Specific Heat Capacity (kJ/kg-°C)
4.18700

Latent Heat of Condensation (kJ/kg)
0.00000

Cold Fluid Specific Heat Capacity (kJ/kg-°C)
4.00000

Latent Heat of Vaporisation (kJ/kg)
0.00000

Heat Transfer Area (A) Calculation

How do you want to provide the Heat Transfer Area (A) m^2?

- Calculate A (Plate Geometry)
- Input A Manually

Enter Effective Plate Surface Area (A in m^2)
170.00

Final Calculated Heat Transfer Area (A): 170.0000 m²

How do you want to provide the Correction Factor (F)?

- (•) Default PHE Factor (F = 0.95)
- ( ) Calculate F (Multi-pass PHE)
- ( ) Input F Manually

Assessment Results

	Parameters	Units	Operation Data	Design Data
0	Heat Exchanger Type	-	Plate Heat Exchanger (PHE)	Plate Heat Exchanger (PHE)
1	Duty (Q)	kW	4972.06	6139.53
2	Hot Fluid Pressure Drop ΔPh	bar	1.20	0.65
3	Cold Fluid Pressure Drop ΔPc	bar	1.30	0.98
4	Temp Range Hot Fluid	°C	17.10	19.10
5	Temp Range Cold Fluid	°C	11.00	16.90
6	Corrected LMTD	°C	9.25	5.00
7	Heat Transfer Coefficient U	kW/m²-K	3.163	7.235

Effectiveness (ε)
0.7095

Capacity Ratio (r)
0.9024

Fouling Resistance (Rd)
0.1779 m²K/kW

AI Performance Diagnostic Summary

Generate AI Assessment Summary

Diagnostic Summary

- **Fouling Margin Assessment:** Current $U_{op} = 3.163\,kW\,m^{-2}\,K^{-1}$ vs. design $U_d = 7.235\,kW\,m^{-2}\,K^{-1}$ (≈ 56 % drop). The 15 % design fouling allowance is fully exhausted; performance is well below the allowable limit.
- **Hydraulic Comparison:** Hot-side $\Delta P = 1.20$ bar (≈ 85 % above design 0.65 bar) and cold-side $\Delta P = 1.30$ bar (≈ 33 % above design 0.98 bar), indicating significant flow restriction and possible channel blockage.
- **Maintenance Recommendation:** Immediate remedial action is required – perform a CIP/chemical cleaning of the plate bundle, followed by visual plate inspection and replacement of any damaged or heavily fouled plates. Re-balance flow distribution before returning to service.

Brief Engineering Insight

The steep decline in U ($U_{op} \approx 3.16\,kW\,m^{-2}\,K^{-1}$) together with a high calculated fouling resistance ($R_d \approx 0.178\,m^2\,K\,kW^{-1}$) raises the overall thermal resistance, forcing the exchanger to operate at a larger corrected LMTD (9.25 °C) to meet duty, which depresses the effectiveness to 0.71 despite a near-balanced capacity ratio ($r \approx 0.90$). The concomitant rise in pressure drops reflects reduced hydraulic cross-section from fouling/scaling, confirming that both heat-transfer and flow-path degradation are driving the observed performance loss.